

Sai Sumedh Kaveti

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EDUCATION

North Carolina State University

Raleigh, NC

Master of Science in Computer Science; **GPA: 3.72/4.0**

Expected May 2027

Relevant Coursework: Design and Analysis of Algorithms, Software Engineering, DevOps: Modern Software Engineering Practices, Computer Networks, Computer and Network Security, Human-Computer Interaction, Game Engine Foundations, Automated Learning and Data Analysis, Neural Networks

VNR Vignana Jyothi Institute of Engineering and Technology

Telangana, India

Bachelor of Technology in Computer Science – Data Science; **GPA: 8.93/10.0**

Dec 2021 – May 2025

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java, C++, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, SQLite

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, CI/CD

Engineering Concepts: System Design, REST APIs, Webhooks, Authentication, Authorization, Asynchronous Processing, Concurrency, Idempotency

Tools & Practices: Git, GitHub, Agile/Scrum, Automated API Testing

Data/ML: Pandas, NumPy, scikit-learn

PROJECTS

TaskFlow – Team Task Management Platform | TypeScript, React, Node.js, Express.js, PostgreSQL, Docker

- Built a full-stack team task management platform using **TypeScript, React, Node.js, Express.js, and PostgreSQL** with project creation, member management, task assignment, priorities, due dates, and Kanban-style status tracking.
- Implemented **JWT authentication, role-based project authorization, protected REST APIs, search and filtering, and relational PostgreSQL models** to securely manage project members, tasks, and ownership permissions.
- Added **automated API testing, Docker-based application packaging, and GitHub Actions CI**, and deployed the frontend, backend, and PostgreSQL services as a publicly accessible full-stack application.

FinSight – Personal Finance Dashboard | TypeScript, React, Node.js, Express.js, PostgreSQL, Docker

- Developed a full-stack personal finance platform using **TypeScript, React, Node.js, Express.js, and PostgreSQL** for managing financial accounts, recording transactions, and tracking monthly budgets.
- Designed **PostgreSQL-backed financial analytics and REST APIs** to calculate account balances, category-level spending, budget utilization, monthly cash flow, and multi-month spending trends.
- Implemented **JWT authentication, per-user data isolation, transaction search and filtering, 6 API integration tests, Docker containerization, and GitHub Actions CI** for reliable development and automated validation.

InsightChat AI – AI-Powered Data Analysis Platform | TypeScript, React, Node.js, Express.js, PostgreSQL, OpenAI API

- Built a full-stack data analysis platform that allows users to **upload CSV datasets, inspect structured data, and explore automatically generated statistics and interactive visualizations** through a React dashboard.
- Integrated the **OpenAI API** to provide a contextual AI assistant that answers natural-language questions about uploaded datasets, summarizes trends, identifies key patterns, and generates data-driven insights.
- Developed **JWT-secured REST APIs, PostgreSQL-backed dataset and conversation history, CSV parsing and validation, automated integration tests, Docker containerization, and GitHub Actions CI**.

EXPERIENCE

North Carolina State University, Poole College of Management

Raleigh, NC

Graduate Teaching Assistant

Aug 2026 – Present

- Support **162 students across three course sections** through **12–15 hours of weekly Python and SQL debugging support**, diagnosing implementation errors and explaining technical concepts.
- Review programming and analytics assignments, identify recurring implementation errors, and provide structured technical feedback while maintaining consistent evaluation standards.

Indian National Centre for Ocean Information Services (INCOIS)

Telangana, India

Machine Learning Intern

Oct 2024 – Apr 2025

- Improved climate forecast model accuracy by **18%** by engineering an end-to-end **Python regression pipeline** over 100+ years of global meteorological data.
- Reduced data-processing time by **30%** by automating preprocessing workflows and organizing century-scale temperature processing into reusable Python components.